User Based Recommendation

Function Explanation:

def predict(ratings, similarity, type='user'):

Creating function with parameter passed.
Parameters:
(i) ratings → User-item rating matrix
(ii) similarity → Similarity matrix
(iii) type='user' → Chooses prediction method:

if type == 'user':

This block predicts ratings based on similar users.

mean_user_rating = ratings.mean(axis=1).to_numpy()

Calculate average rating of each user:

axis=1 means row-wise average.

.to_numpy() converts pandas Series into NumPy array.

ratings_diff = ratings - mean_user_rating[:, np.newaxis]

Normalize ratings:

This subtracts each user's average rating from their ratings.
Normalization removes personal bias.

pred = mean_user_rating[:, np.newaxis] + \similarity.dot(ratings_diff) / \ np.array([np.abs(similarity).sum(axis=1)]).T

Predict Ratings:

Predicted rating = User average rating + weighted ratings from similar users

elif type == 'item':

This predicts ratings using similar items.

pred = ratings.dot(similarity) / np.array([np.abs(similarity).sum(axis=1)])

Predicted rating depends on:

- ratings of similar items
- weighted by item similarity

return pred - Returns predicted ratings matrix.